

Generation Interconnection System Impact Study Report

For Queue Project Q11-0069

West Bassett 138 kV

February 22, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0069. The Interconnection Customer proposes a 49.7 MW solar photovoltaic facility interconnecting at the West Bassett 138 kV substation in the Virginia area. The request was received on September 07, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 59 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,939,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Beardsley St-Red Hills 138kV	Bear Substation-Clover 345kV	119.7	234	8.0
Vaseaux Lake-Red Hills 138kV	Bear-Clover 345kV	112.8	239	8.0
Mcclain Energy Facility-Vaseaux Lake Substation 138kV	Bear-Clover 345kV Line	111.2	240	8.0
Hutchins-Bauxite 138kV	Bear-Clover 345kV	110.5	329	5.0
Lone Tree-Craven County Wood Energy Lp 138kV	Bear-Clover 345kV line	107.7	226	6.9
Boonsboro-Nacoochee 138kV	Bear-Clover 345kV	106.5	354	6.2
Trinidad-State School 138kV Line	Mecklenburg Power Station-Person 345kV	106.3	419	5.5
Franklin-Elk Grove-S Texa 138kV Line	S Texa-Desert Outpost 138kV	106.3	239	6.0
Nacoochee-S Texa 138kV	Bear-Clover 345kV	106.2	358	6.2
Craven County Wood Energy Lp-Lowmoor 138kV	Bear-Clover 345kV	103.8	230	6.9
State School Substation 345-138 kV Autotransformer	Mecklenburg Power Station-Person Substation 345kV	103.5	501	5.5
Clearbrook-Jack Ramey Wind 138kV	Floyds Knobs-Poages Mill 138kV	102.8	296	9.3
Lakeshore-Fresh Meadows 138kV	Bear-Clover 345kV L	100.6	172	5.7
Highway 59-Lakeshore 138kV	Bear-Clover 345kV	100.4	188	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Beardsley St-Red Hills 138kV	Rebuild line with higher-capacity conductor	\$11,251,000	3% (\$293,000)
Vseaux Lake-Red Hills 138kV	Rebuild line with higher-capacity conductor	\$16,090,000	3% (\$419,000)
Mcclain Energy Facility-Vaseaux Lake 138kV	Rebuild line with higher-capacity conductor	\$11,218,000	3% (\$292,000)
Hutchins-Bauxite 138kV	Rebuild line with higher-capacity conductor	\$3,509,000	100% (\$3,509,000)
Lone Tree-Craven County Wood Energy Lp 138kV	Reconductor line	\$5,472,000	2% (\$129,000)
Boonsboro-Nacoochee 138kV	Reconductor line	\$2,184,000	5% (\$101,000)
Trinidad-State School 138kV	Reconductor line	\$1,515,000	100% (\$1,515,000)
Franklin-Elk Grove-S Texa 138kV Line	Reconductor line	\$3,272,000	3% (\$94,000)
S Texa-Nacoochee 138kV	Reconductor line	\$1,064,000	5% (\$49,000)
Craven County Wood Energy Lp-Lowmoor 138kV	Reconductor line	\$5,942,000	2% (\$140,000)
State School Substation 345/138kV XFMR	Replace transformer with higher-rated unit	\$18,752,000	100% (\$18,752,000)
Clearbrook-Jack Ramey Wind 138kV	Reconductor line	\$703,000	4% (\$26,000)
Lakeshore-Fresh Meadows 138kV	Reconductor line	\$3,234,000	100% (\$3,234,000)
Highway 59-Lakeshore 138kV	Reconductor line	\$1,386,000	100% (\$1,386,000)
	Total allocated to project		\$29,939,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.